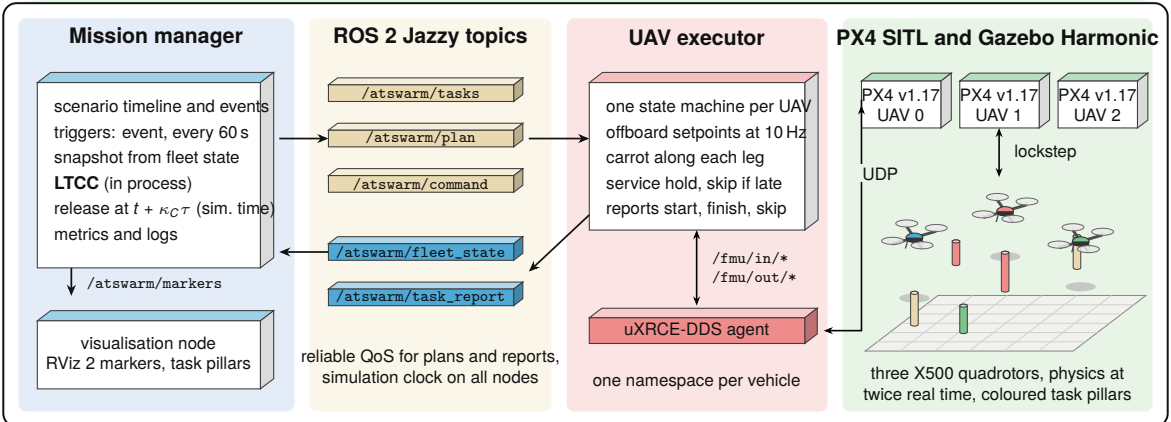

rescale
$S_X = 0.1$, $S_T = 0.25$
speed $\times S_X / S_T$,
energy $\times S_X$
$\kappa_C = S_T\,\kappa_B$
utility per flight, compared with the same mission in tier B
Tier B: 10 km, 1800 s
same dimensionless ratios
Tier C: 1 km, 450 s, 3 UAVs
same seeds, same disturbances


Mission manager
scenario timeline and events
triggers: event, every 60 s
snapshot from fleet state
LTCC (in process)
release at $t + \kappa_C \tau$ (sim. time)
metrics and logs
/atswarm/markers
visualisation node
RViz 2 markers, task pillars
ROS 2 Jazzy topics
/atswarm/tasks
/atswarm/plan
/atswarm/command
/atswarm/fleet_state
/atswarm/task_report
reliable QoS for plans and reports, simulation clock on all nodes
UAV executor
one state machine per UAV
offboard setpoints at 10 Hz
carrot along each leg
service hold, skip if late
reports start, finish, skip
/fmu/in/*
/fmu/out/*
uXRCE-DDS agent
one namespace per vehicle
PX4 SITL and Gazebo Harmonic
PX4 v1.17 UAV 0
PX4 v1.17 UAV 1
PX4 v1.17 UAV 2
lockstep
UDP
three X500 quadrotors, physics at twice real time, coloured task pillars